

first industrial robot. And in 1969 was invented the Stanford Arm, the first electrically-powered, computer-controlled robot arm. Talking about robotics, where was AI? Well, it was there, and the research being done back then doesn't seem primitive compared to what's being done today. OK, a lot has been done since then, but think about how much we've done and how far we need to go. They were developing chat-bots and neural networks back then, and we're developing chat-bots and neural networks now as well.

Still, the point is that a lot of work done back then was seminal. AI was in vogue. Frank Rosenblatt built, in 1960, the Perceptron—a sort of computer that could actually learn by trial and error. It employed a neural network. (Neural networks themselves were already in existence.) And then in 1966 happened Eliza, probably the most famous chat-bot of all time. Joseph Weizenbaum developed Eliza at MIT: “she” was a psychiatrist, and would chat with people primarily in consulting mode. “Tell me more about your mother.” “Does that make you feel depressed?” You can find a Web version of the original Eliza at www-ai.ijs.si/eliza/eliza.html.

Why was Eliza important? Mostly because it proved that something like a chat-bot could be built; people were fascinated by the fact that a computer seemed to be exhibiting intelligence.

More in AI: a team at Stanford led by Ed Feigenbaum created DENDRAL, the first expert system. In an expert system, one feeds in as much knowledge in a certain domain as one can into the system, and it analyses the situation and provides answers comparable in quality to what a human expert would propose. What DENDRAL

did is not very interesting, but expert systems had come into being. They're used today in many domains, but medicine is one field that has particularly benefited.

Come to think of it, a lot happened in the '60s. Did you think the CD was invented by Philips in the '80s? Not really: the man behind the CD (and therefore, optical storage) was a less-known-today inventor called James Russell. Actually, he invented a complete optical digital recording system, not just the storage medium. In an interview, asked what prompted him to explore the concept of optical recording, Russell's answer was that he developed the technology because he “wanted a better hi-fi system”! Talk about necessity being the mother of...

We won't bore you with details of what research papers were written around this time. But so much tech that we take for granted started off as single research papers! At Rand Corporation, Paul Baran developed, in 1960, the packet-switching principle. This was followed, in 1961, by the first paper on packet-switching theory, by Leonard Kleinrock at MIT: it was called “Information Flow in Large Communication Nets.” What talk of the Internet without packet switching?

Coming to the Internet, the first plans for ARPANET were laid out by Larry Roberts, also of MIT, in a 1966 paper called “Towards a Co-operative Network of Time-Shared Computers.” ARPANET, as you might know, was what eventually “became the Internet.” We'll be talking more, as we should, about ARPANET—it's not as simple as “it was the first Internet.” In fact, some of the foundations of the Internet were laid in the 60s: way back in 1960, J C R Licklider

published a seminal paper called “Man-Computer Symbiosis.” It included these excerpts:

“It seems reasonable to envision, for a time 10 or 15 years hence, a ‘thinking center’ that will incorporate the functions of present-day libraries together with anticipated advances in information storage and retrieval.

“The picture readily enlarges itself into a network of such centers, connected to one another by wide-band communication lines and to individual users by leased-wire services...”

Sound sufficiently enough like something we're familiar with today?

To avoid hate-mail for having missed out on this point, we must mention that Unix was developed by Ken Thompson and Dennis Ritchie in 1969 at Bell Labs, the same year Linus Torvalds was born. We don't need to tell you how important that development was.

Yes, much was done in the 1960s, but we think the seventies were even more exciting—and something we can relate to better! Turn the page! ■

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DEC's PDP-1

There are many, many firsts in the world of personal computing; the DEC PDP-1 is just one of those firsts, but it sounds suspiciously like our PCs. It's been described as the “first commercial computer with a monitor and keyboard input,” as the “world's first small, interactive computer,” and so on. Essentially, this was the first thing that felt—if not looked—something like what we all have on our desks today. It was an 18-bit machine with a keyboard and a light pen for input. It was small—about the size of a couple of fridges. One of its biggest pluses was that it didn't require specialised air conditioning, and that it could run on standard (US) 110-volt power. Some PDP-1s had CRT graphics display systems, which happened to be the first commercially-available computer graphics terminals.

A PDP-1 at MIT became the first computer on which a game was played: Spacewar!, which became a DEC standard diagnostic program for the rest of the decade.

